

AI-Powered Security Operations Center (SOC): Architecture Overview

1. SOC Challenges & AI Augmentation

Modern enterprise SOC's face alert fatigue with 10,000+ daily alerts, 70% of which are false positives. An AI SOC pipeline automates anomaly detection and incident report generation, reducing Mean Time to Detect (MTTD) from days to minutes.

2. AI SOC Pipeline Architecture

Pipeline Stage	Technology	Functionality
Telemetry Ingestion	SIEM / Elastic / Kafka	Ingests firewall, authentication, and endpoint logs
ML Anomaly Detection	Isolation Forest / Autoencoder	Detects statistically anomalous traffic and authentication spikes
Alert Clustering & Scoring	DBSCAN / Graph Analytics	Groups related security events into unified incident contexts
Generative Incident Reporter	GPT-4 / Llama 3 Security	Produces structured MITRE ATT&CK; mapped incident summaries
Analyst Triage Interface	Web Dashboard + Slack Alerts	Enables 1-click incident containment and firewall rule deployment